

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing techniques. (BL4,BL5)
Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:40

Annexure A

MOCK INTERVIEW

Code of Conduct:

*I Naveen Kumar V V (192321018)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
naveenkumarvv

Annexure A

Mock Interview – Sample Questions (5 Questions)

Mock Interview Test Format for Object Oriented Analysis and Design (OOAD)

Section A	Self Introduction & Fundamentals	3 Minutes	5
Section B	Technical Questions (OOAD Concepts)	7 Minutes	10
Section C	UML Modeling & Design	7 Minutes	10
Section D	Case Study / Problem Solving	8 Minutes	10
Section E	HR & Career-Oriented Questions	5 Minutes	5

SECTION A – SELF INTRODUCTION (5 Marks)

Evaluation Criteria

- Communication Skills
- Confidence
- Technical Awareness
- Academic Background

Sample Questions

1. Tell me about yourself.
 2. Why did you choose Computer Science?
 3. Explain your favorite subject in OOAD.
 4. What software development tools have you used?
 5. Describe a project where you applied object-oriented concepts.
-

SECTION B – TECHNICAL QUESTIONS (15 Marks)

Basic Level

Q1. What is Object-Oriented Analysis and Design?

Q2. Differentiate between Analysis and Design.

Q3. Explain Encapsulation with an example.

Q4. What is Inheritance? Mention its types.

Q5. Define Polymorphism and its benefits.

Q6. What is Abstraction?

Q7. Difference between Aggregation and Composition.

Q8. What is Coupling and Cohesion?

Q9. What is a Class and an Object?

Q10. Explain Object Responsibility.

SECTION C – UML MODELING (15 Marks)

Q11. What is UML?

Q12. When would you use a Use Case Diagram?

Q13. Draw and explain a Class Diagram for a Library Management System.

Q14. Differentiate Sequence Diagram and Activity Diagram.

Q15. Explain State Diagram with an example.

Q16. What is a Deployment Diagram?

Q17. How do Component Diagrams help software architects?

Q18. What is Multiplicity in UML?

SECTION D – CASE STUDY / PROBLEM SOLVING (10 Marks)

Case Study 1

A hospital wants a system to manage:

- Patients
- Doctors
- Appointments
- Billing

Questions

1. Identify the classes.
 2. Draw a simple class diagram.
 3. Suggest relationships among classes.
 4. Which UML diagrams would you use for analysis?
 5. Explain how inheritance can be applied.
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Case Study 2

An Online Shopping System supports:

- Customers
- Products
- Cart
- Orders
- Payments

Questions

1. Identify actors and use cases.
 2. Design a use case diagram.
 3. Suggest suitable design patterns.
 4. Explain how polymorphism can be used in payment processing.
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SECTION E – HR & PROFESSIONAL QUESTIONS (5 Marks)

Q19.Why should we hire you?

Q20.What are your strengths and weaknesses?

Q21.How do you handle project deadlines?

Q22.Have you worked in a team project? Explain your role.

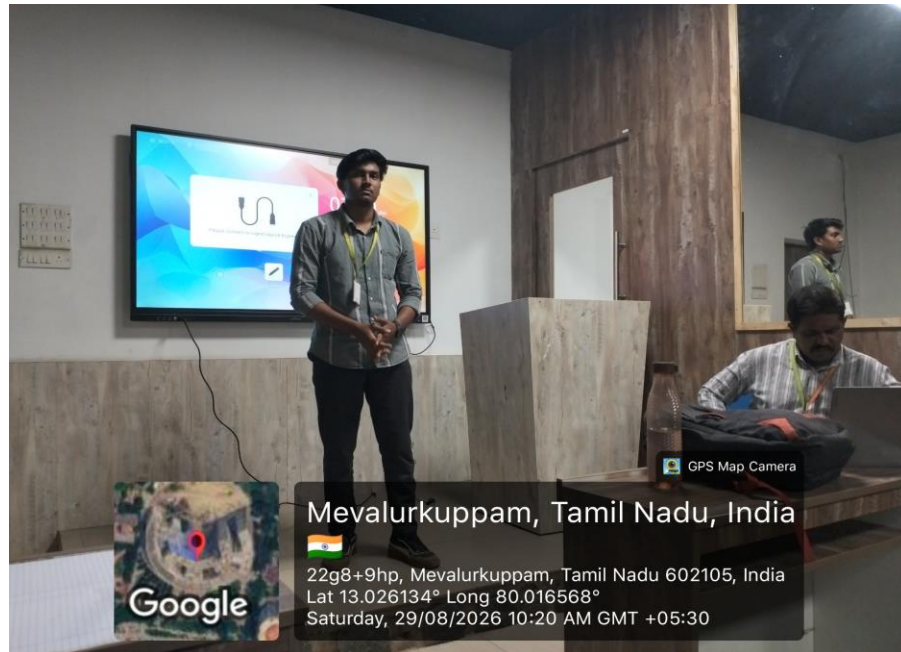
Q23.Where do you see yourself in five years?

RUBRICS FOR CALCULATION:**MOCK INTERVIEW**

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

STUDENT PHOTOGRAPH

V V Naveen Kumar | Reg. No. 192321018 | Information Technology (IT)



Student Photo

ANNEXURE A**MOCK INTERVIEW – COMPLETED SUBMISSION**

Section	Area	Time	Marks
A	Self Introduction & Fundamentals	3 Minutes	5
B	Technical Questions (OOAD Concepts)	7 Minutes	10
C	UML Modeling & Design	7 Minutes	10
D	Case Study / Problem Solving	8 Minutes	10
E	HR & Career-Oriented Questions	5 Minutes	5

SECTION A – SELF INTRODUCTION (5 Marks)

Evaluation criteria: Communication Skills, Confidence, Technical Awareness, and Academic Background.

1. Tell me about yourself.

Answer: My name is V V Naveen Kumar and my register number is 192321018. I am a student of Information Technology with a strong interest in software development, artificial intelligence, data science, and object-oriented design. I have worked with Python, C++, Java, MySQL, HTML, CSS and JavaScript, and I am familiar with tools such as VS Code, GitHub, Figma and Canva. I enjoy solving practical problems and building technology-based solutions.

2. Why did you choose Computer Science?

Answer: I chose Computer Science because it combines logical thinking, creativity and problem solving. It provides opportunities to develop software that can solve real-world problems and allows me to continuously learn new technologies.

3. Explain your favorite subject in OOAD.

Answer: My favorite part of OOAD is UML modeling because it helps convert requirements into a clear software design. Use case, class and sequence diagrams make system responsibilities and interactions easier to understand before implementation.

4. What software development tools have you used?

Answer: I have used Visual Studio Code for development, Git and GitHub for version control, MySQL for database work, Figma and Canva for design, and AI-assisted development tools such as ChatGPT and GitHub Copilot.

5. Describe a project where you applied object-oriented concepts.

Answer: In a software project, I applied classes and objects to represent real-world entities, used encapsulation to protect data, and used abstraction to separate essential behavior from implementation details. This approach made the system modular, easier to test, maintain and extend.

SECTION B – TECHNICAL QUESTIONS (OOAD)**Q1. What is Object-Oriented Analysis and Design?**

Answer: OOAD is a software engineering approach that models a system as interacting objects. Object-Oriented Analysis identifies requirements, objects, actors and responsibilities, while Object-Oriented Design converts those findings into classes, interfaces, relationships and implementation-level structures.

Q2. Differentiate between Analysis and Design.

Answer: Analysis focuses on understanding what the system must do and identifying requirements, actors, objects and responsibilities. Design focuses on deciding how the system will do it by defining classes, interfaces, relationships, algorithms and architecture.

Q3. Explain Encapsulation with an example.

Answer: Encapsulation bundles data and methods inside a class and restricts direct access to internal data. For example, a BankAccount class can keep balance private and provide deposit() and withdraw() methods to control how the balance is changed.

Q4. What is Inheritance? Mention its types.

Answer: Inheritance allows a child class to acquire properties and behavior from a parent class. Common types are single, multilevel, hierarchical, multiple and hybrid inheritance. In Java, multiple inheritance of classes is not supported directly, but it can be achieved through interfaces.

Q5. Define Polymorphism and its benefits.

Answer: Polymorphism means one interface or operation can have different implementations. It includes compile-time polymorphism such as method overloading and run-time polymorphism through method overriding. It improves flexibility, extensibility and code reuse.

Q6. What is Abstraction?

Answer: Abstraction hides unnecessary implementation details and exposes only essential features. For example, a Payment interface may expose pay() while hiding the internal processing used by card, UPI or wallet payments.

Q7. Difference between Aggregation and Composition.

Answer: Aggregation is a weak whole-part relationship where the part can exist independently of the whole, such as a Department and Teacher. Composition is a strong ownership relationship where the part normally depends on the whole, such as a House and its Rooms.

Q8. What is Coupling and Cohesion?

Answer: Coupling measures the dependency between modules; low coupling is preferred because changes in one module have less impact on others. Cohesion measures how closely related the responsibilities within a module are; high cohesion is preferred because each module has a focused purpose.

Q9. What is a Class and an Object?

Answer: A class is a blueprint that defines attributes and behaviors. An object is a runtime instance of a class. For example, Car is a class, while myCar can be an object created from that class.

Q10. Explain Object Responsibility.

Answer: Object responsibility is the work an object is expected to perform or the information it is responsible for maintaining. Assigning responsibilities clearly improves cohesion and prevents one class from becoming a large, difficult-to-maintain component.

SECTION C – UML MODELING**Q11. What is UML?**

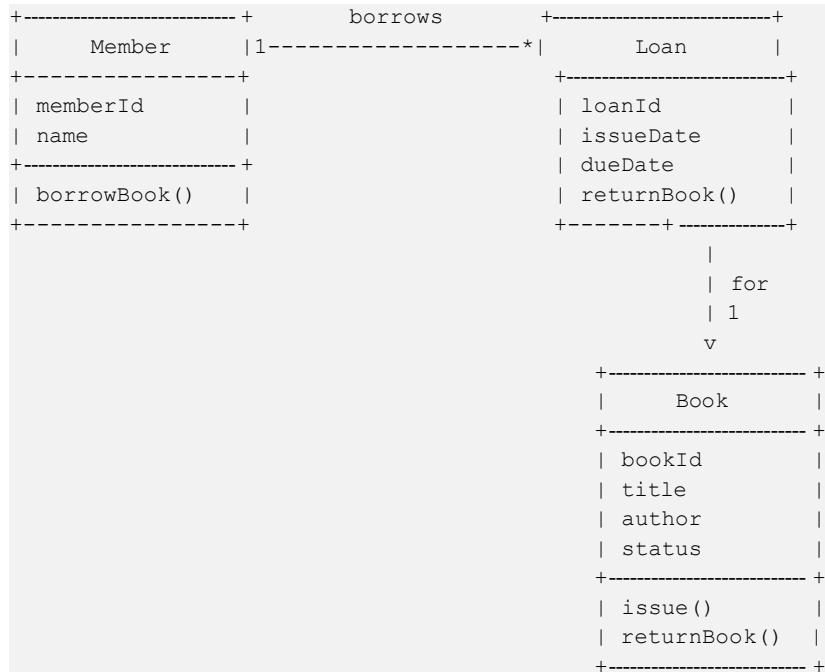
Answer: UML (Unified Modeling Language) is a standardized visual language used to specify, visualize, construct and document software systems. It provides diagrams such as use case, class, sequence, activity, state, component and deployment diagrams.

Q12. When would you use a Use Case Diagram?

Answer: A use case diagram is used during requirements analysis to show system actors and the functions they expect from the system. It gives a high-level view of system scope and user interactions.

Q13. Draw and explain a Class Diagram for a Library Management System.

Answer: A simple class model can contain Member, Book, Librarian and Loan. Member borrows books through Loan, while Librarian manages books and loans.



Librarian manages Books and Loans.

Relationships: Member-Loan (1:M), Loan-Book (M:1), Librarian-Book (1:M).

Explanation: The class diagram identifies the main entities, their attributes, operations and relationships. Multiplicity shows how many objects participate in each relationship.

Q14. Differentiate Sequence Diagram and Activity Diagram.

Answer: A sequence diagram emphasizes the time-ordered messages exchanged between objects or components. An activity diagram emphasizes workflow, actions, decisions and parallel activities in a process.

Q15. Explain State Diagram with an example.

Answer: A state diagram represents the different states of an object and the events that cause transitions. For an Order object, possible states are Placed → Confirmed → Shipped → Delivered. A cancellation event may move an eligible order to Cancelled.

Q16. What is a Deployment Diagram?

Answer: A deployment diagram shows the physical arrangement of hardware or execution nodes and the software artifacts deployed on them. For a web application it can show a client device, web server, application server and database server.

Q17. How do Component Diagrams help software architects?

Answer: Component diagrams show the major software components, their provided/required interfaces and dependencies. They help architects understand modular structure, integration points, replaceable components and system dependencies.

Q18. What is Multiplicity in UML?

Answer: Multiplicity specifies how many instances of one class can be associated with an instance of another class. Common values include 1, 0..1, 1..*, and 0..*. For example, one customer may place many orders, represented as Customer 1 — 0..* Order.

SECTION D – CASE STUDY / PROBLEM SOLVING

Case Study 1: A hospital system manages Patients, Doctors, Appointments and Billing.

1. Identify the classes.

Answer: Patient, Doctor, Appointment and Bill/Billing are the main classes. Additional classes such as Receptionist or MedicalRecord can be introduced if the system scope requires them.

2. Draw a simple class diagram.

3. Suggest relationships among classes.

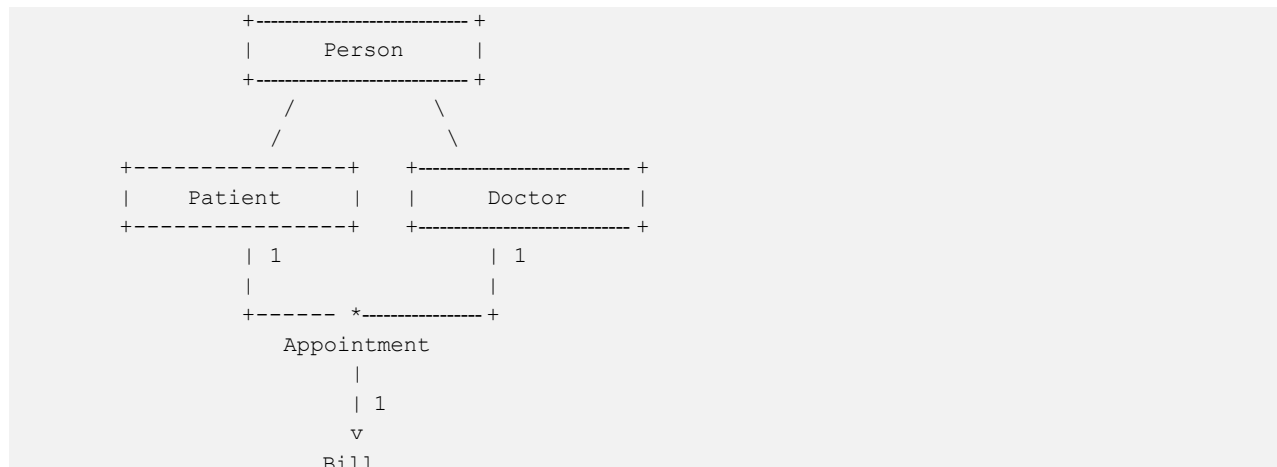
Answer: A Patient can have many Appointments; a Doctor can attend many Appointments; an Appointment connects one Patient with one Doctor. A Patient can have billing records, and a Bill is associated with the relevant appointment or patient.

4. Which UML diagrams would you use for analysis?

Answer: Use Case Diagram for actors and requirements, Class Diagram for structural entities and relationships, Activity Diagram for hospital workflows, Sequence Diagram for interactions such as booking an appointment, and State Diagram for appointment states.

5. Explain how inheritance can be applied.

Answer: A general Person class can contain common attributes such as personId, name and contact. Patient and Doctor can inherit from Person and add specialized attributes and operations. This reduces duplication while keeping role-specific behavior separate.



Case Study 1 conclusion: The design separates patient, doctor, appointment and billing responsibilities, which supports low coupling, high cohesion and easier maintenance.

Case Study 2: An Online Shopping System supports Customers, Products, Cart, Orders and Payments.

1. Identify actors and use cases.

Answer: Actors include Customer, Payment Service and optionally Administrator. Customer use cases include Register/Login, Browse Products, Add to Cart, Update Cart, Place Order, Make Payment, Track Order and View Order History. Administrator can manage products and orders.

2. Design a use case diagram.

3. Suggest suitable design patterns.

Answer: Factory Method or Abstract Factory can create different payment processors; Strategy can encapsulate different payment algorithms; Observer can support order-status notifications; Repository can separate persistence logic from business logic.

4. Explain how polymorphism can be used in payment processing.

Answer: Define a common Payment interface with pay(amount). CardPayment, UpiPayment and WalletPayment implement the same interface differently. The checkout service can call payment.pay(amount) without depending on the concrete payment type.

```
Customer ---> (Browse Products)
Customer ---> (Add to Cart)
Customer ---> (Place Order) ---> (Make Payment)
Customer ---> (Track Order)
Customer ---> (View Order History)

Payment Service <--- (Make Payment)

Administrator ---> (Manage Products)
Administrator ---> (Manage Orders)
```

Use-case explanation: The diagram keeps the customer-facing functions separate from external payment processing and administrative functions, making system scope and responsibilities clear.

SECTION E – HR & PROFESSIONAL QUESTIONS (5 Marks)

Q19. Why should we hire you?

Answer: You should hire me because I have a strong willingness to learn, a problem-solving mindset and practical exposure to programming, databases, software tools and project development. I can work independently as well as in a team, and I am committed to improving my technical and professional skills.

Q20. What are your strengths and weaknesses?

Answer: My strengths are quick learning, logical problem solving, communication, decision making and teamwork. One weakness is that I sometimes spend extra time refining details; I am improving this by prioritizing tasks and setting clear time limits.

Q21. How do you handle project deadlines?

Answer: I divide the project into smaller tasks, prioritize work according to importance, estimate the required time and track progress regularly. I communicate early when I identify a risk and focus first on completing critical functionality.

Q22. Have you worked in a team project? Explain your role.

Answer: Yes. In team projects, I have contributed to planning, development, testing, documentation and coordination. I focus on completing my assigned module, integrating it with other modules and communicating issues clearly so that the team can meet its common goal.

Q23. Where do you see yourself in five years?

Answer: In five years, I see myself as a skilled software professional who can design, develop and maintain reliable applications. I want to strengthen my expertise in software engineering, AI and data-driven systems while taking greater responsibility in technical projects.

RUBRICS FOR CALCULATION – MOCK INTERVIEW

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Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding
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Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples

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